

# Experiment 6

## 6.1.2 Factorial of a number

### Algorithm :

Step 1 : Start

Step 2 : Input n

Step 3 :  $i = 1$

fact = 1

Step 4 : if (  $i \leq n$  ) go to step 5

Step 5 : fact  $\times = i$

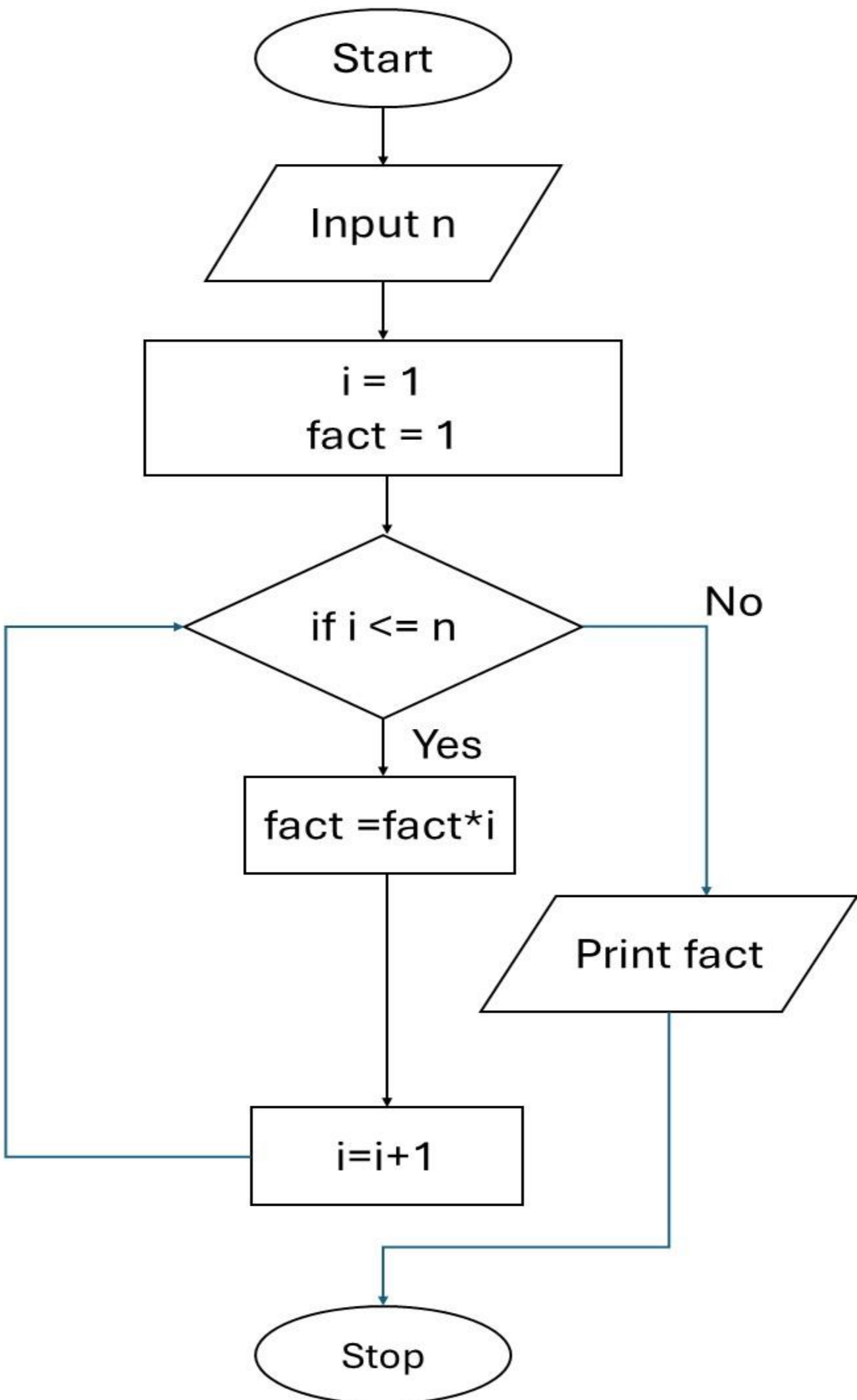
Step 6 :  $i = i + 1$

Step 7 : Go to step 4

Step 8 : Print fact

Step 9 : Stop

# Flowchart :



# Code :

```
n=int(input())
fact = 1
i=1
for i in range(1,n+1):
    fact*=i
print(fact)
```

# Execution :

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6.1.2. Factorial of a Number

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Write a Python program to calculate the factorial of a number  $n$  using loops.

Input Format:

- A single line containing an integer  $n$ .

Output Format:

- Print the factorial of the given integer  $n$ .

Sample Test Cases

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factorialN...

1 n=int(input())

2 fact = 1

3 i=1

4 for i in range(1,n+1):

5 fact\*=i

6 print(fact)

Submit

Average time

0.005 s

5.25 ms

Maximum time

0.007 s

7.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 7 ms

Test case 2 5 ms

Terminal

Test cases

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